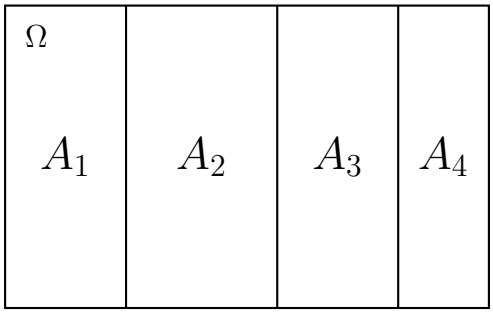
Check-in:

- Universal set Ω
- Mutually exclusive (disjoint) sets $A \cap B = \emptyset$
- Complement of a set A A^C
- Difference of two sets $A \& B$ $A - B$

Partition: A partition U is a collection of disjoint subsets $A_i \subset \Omega$ whose union equals Ω.

This means

- $U = \{A_1, A_2, \cdots, A_n\}$
- $A_i \cap A_j = \emptyset, \forall i \neq j$ "pairwise" disjoint
- $\Omega = A_1 \cup A_2 \cup \cdots \cup A_n$



(e.g., n=4)

$U = \{A_1, A_2, A_3, A_4\}$ is a partition of Ω.

Finite, Countable, and Uncountable Sets

- **Finite set:** A set S is finite if it contains a finite number of elements. The **cardinality** of a finite set S, denoted by $|S|$, is the number of elements in S.

  For example,

  $$S = \{1, 2, 3, 4, 5, 6\} \qquad \text{and} \qquad |S| = 6.$$

- **Countably infinite set:** A set S is countably infinite if its elements can be put into one-to-one correspondence with the positive integers

  $$\mathbb{N} = \{1, 2, 3, \ldots\}.$$

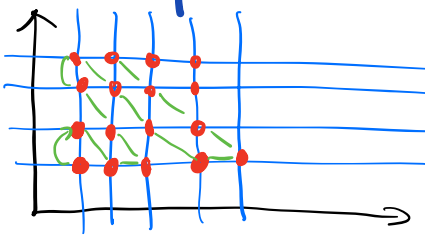
  In other words, its elements can be listed as

  $$S = \{s_1, s_2, s_3, \ldots\}.$$

  For example,

  $$S = \{0, 1, 2, 3, \ldots\}.$$

Example. The set of rational numbers is countably infinite.
Any rational number can be written as $\frac{m}{n}$ where
$m \in \mathbb{Z}$ (integer set), $n \in \mathbb{N} = \{1, 2, 3, \cdots\}$. We can
think of each rational number $\frac{m}{n}$ as a pair (m, n)



We can count by following a zig-zag trajectory.
Simply skip repetitions.
We list all positive rationals:
$q_1, q_2, q_3, \cdots$

Then, we can count all rationals as follows:

$0, q_1, -q_1, q_2, -q_2, q_3, -q_3, \cdots \cdots$

- **Countable set:** A set is countable if it is either finite or countably infinite.

- **Uncountable set:** A set is uncountable if its elements cannot be listed in this way.

  For example,

  $$S = [0, 1]$$

  is uncountable.

Applying Set Theory to Probability

Example

Suppose a fair six-sided die is rolled once. What is the probability that the number rolled is less than 4?

- **Experiment:** the procedure being performed - rolling a fair six-sided die once.
- **Outcome:** one possible result, e.g., 3.
- **Sample space:** the set of all possible outcomes,
$$S = \{1, 2, 3, 4, 5, 6\}.$$
- **Event:** a set of outcomes of interest. For "less than 4,"
$$A = \{1, 2, 3\}.$$
- **Probability:** measure the likelihood of the event,
$$P(A) = \frac{|A|}{|S|} = \frac{3}{6} = \frac{1}{2}.$$

Basic Probability Terminology

- **Experiment:** An experiment is a process that produces an outcome. The outcome is not known with certainty before the experiment is performed.
- **Outcome:** An outcome is one possible result of an experiment.
- **Sample Space:** The sample space, denoted by S, is the set of all possible outcomes of an experiment.
- **Event:** An event is a subset of the sample space. That is,
$$A \subset S.$$
An event occurs if the outcome of the experiment belongs to A.
- **Probability:** The probability of an event A, denoted by $P(A)$, is a number that measures the likelihood that the event A occurs:
$$0 \leq P(A) \leq 1.$$

Engineering Example: Digital Communication

Suppose a digital communication system receives two bits. Each received bit can be either 0 or 1.

- **Experiment:** Observe the two received bits.
- **Outcome:** One possible received bit pair, e.g.,

$$10.$$

- **Sample Space:** The set of all possible received bit pairs:

$$S = \{00, 01, 10, 11\}.$$

- **Event:** Let A be the event that at least one received bit is 1. Then

$$A = \{01, 10, 11\}.$$

- **Probability:** If the four outcomes are "equally likely," then

$$P(A) = \frac{|A|}{|S|} = \frac{3}{4}.$$

No one outcome is any more likely than any other.

Set Terminology	Probability Terminology
Universal set S	Sample space S
Element $\omega \in S$	Outcome
Subset $A \subset S$	Event A
Empty set $\emptyset$	Impossible event
S	Certain event
A^c	Event "A does not occur"
$A \cup B$	Event "A or B occurs"
$A \cap B$	Event "A and B occur"
$A \cap B = \emptyset$	A and B are mutually exclusive

Probability

Probability assigns a number between 0 and 1 to each event of a sample space, such that the number is a measure of how likely the event is. It is a mapping from a random event to a number between 0 and 1, e.g., $P(A) = 0.1$.

There are different approaches to understanding probability:

- Relative-frequency approach
- Axiomatic approach

Probability:Relative-frequency approach

Suppose an experiment is repeated n times and event A occurs $N_A(n)$ times. The relative frequency of A is

$$f_n(A) = \frac{N_A(n)}{n}.$$

For a large number of repetitions, the relative frequency often stabilizes around the probability of A:

$$P(A) \approx \frac{N_A(n)}{n}$$

and

$$P(A) = \lim_{n\to\infty} \frac{N_A(n)}{n}.$$

$N_A(n)$ n

Example

Toss a coin repeatedly. If heads occurs 503 times in 1000 tosses, its relative frequency is 0.503. As we perform more tosses, we expect this relative frequency to stabilize near the probability of heads.

$$f_n = \frac{N_A(n)}{n} = \frac{503}{1000} = 0.503$$

This gives us an intuitive and empirical interpretation of probability, but we still need a rigorous mathematical framework.

Probability:Axiomatic approach

Rather than defining probability through repeated experiments, the axiomatic approach treats probability as a mathematical function satisfying certain fundamental properties.

Axiomatic Definition of Probability

Let S be the sample space. A *probability measure* P assigns a probability $P(A)$ to each event $A \subset S$ and satisfies the following axioms:

1. **Nonnegativity:**
$$P(A) \geq 0 \qquad \text{for every event } A.$$

2. **Normalization:**
$$P(S) = 1.$$

3. **Countable Additivity:**
If the sequence of events $A_1, A_2, \ldots$ are (pairwise) mutually exclusive, i.e.,
$$A_i \cap A_j = \emptyset, \qquad \forall\, i \neq j,$$
then
$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

$$\Updownarrow$$

$$P(A_1 \cup A_2 \cup \cdots) = P(A_1) + P(A_2) + \cdots \cdots$$

These three axioms are the foundation of probability theory. Many of the probability rules we will use later are consequences of these three axioms. In fact, we will build the entire theory of probability from these three axioms!

Three Axioms of Probability
For any event $A \subset S$:

$$\boxed{1.\ P(A) \geq 0} \qquad \boxed{2.\ P(S) = 1}$$

If $A_1, A_2, \ldots$ are pairwise mutually exclusive, then

$$\boxed{3.\ P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)}$$

Some Properties of Probability

- **Probability of mutually exclusive events**

  Suppose A and B are mutually exclusive:

  $$A \cap B = \emptyset.$$

  By the additivity axiom,

  $$P(A \cup B) = P(A) + P(B).$$

- **Basic consequences of the axioms**

  $$P(\emptyset) = 0,$$

  $$0 \leq P(A) \leq 1,$$

  $$P(A^c) = 1 - P(A).$$

- **Monotonicity**

  If $A \subset B$, then

  $$P(A) \leq P(B).$$

- **General addition rule**

  Suppose A and B are two events that are **not** **necessarily mutually exclusive**, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$